Note that $g(x,3) = 21 - x^2$ represents a parabola; which is obtained by intersecting the surface of $z = 30 - x^2 - y^2$ with the plane $y = 3$ [see Fig. 8.5). Similarly $g(2, y) = 26 - y^2$ represents a parabola; which is obtained by intersecting the surface of $z = 30 - x^2 - y^2$ with the plane $x = 2$ [see Fig. 8.6). Following graphs describes the above discussion.

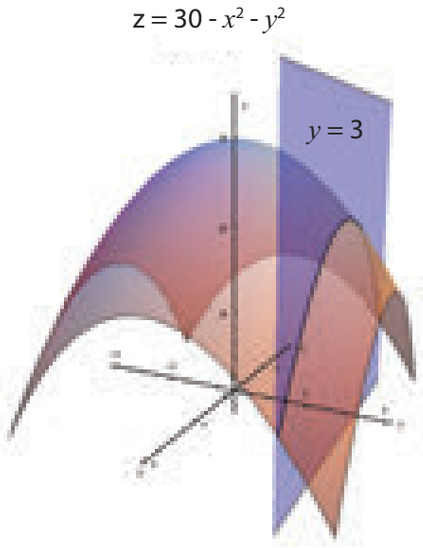


Fig. 8.5

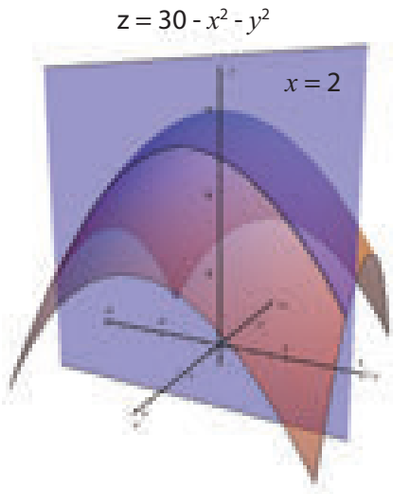


Fig. 8.6

In the same way, given a function F of a two variables say x, y, we can visualize it in the three space by considering the equation $z = F(x, y)$. Generally, this will represent a surface in $\mathbb{R}^3$.

8.3.1 Recall of Limit and Continuity of Functions of One Variable

Next we shall look at continuity of a function of two variables. Before that, it will be beneficial for us to recall the continuity of a function of single variable. We have seen the following definition of continuity in XI Std.

A function $f : (a,b) \to \mathbb{R}$ is said to be continuous at a point $x_0 \in (a,b)$ if the following hold:

(1) f is defined at x_0. (2) $\lim\limits_{x \to x_0} f(x) = L$ exists (3) $L = f(x_0)$

The key idea in the continuity lies in understanding the second condition given above. We write $\lim\limits_{x \to x_0} f(x) = L$ whenever the value $f(x)$ gets closer and closer to L as x gets closer and closer to x_0.

To make it clear and precise, let us rewrite the second condition in terms of neighbourhoods. This will help us when we talk about continuity of functions of two variables.

Definition 8.5 (Limit of a Function)

Suppose that $f : (a,b) \to \mathbb{R}$ and $x_0 \in (a,b)$. We say that f has a limit L at $x = x_0$ if for every neighbourhood $(L - \varepsilon, L + \varepsilon), \varepsilon > 0$ of L, there exists a neighbourhood $(x_0 - \delta, x_0 + \delta) \subset (a,b), \delta > 0$ of x_0 such that

$$f(x) \in (L - \varepsilon, L + \varepsilon) \text{ whenever } x \in (x_0 - \delta, x_0 + \delta) \setminus \{x_0\}.$$

The above condition in terms of neighbourhoods can also be equivalently stated using modulus (or absolute value) notation as follows:

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